Enrollment No:-92400120633

Name:-Smit Hingmire

Class:-4EV3

Lab 5: Develop a multi-page product catalog application implementing routing and Context API-based state management for cart functionality.

Code Snippet:

main.jsx:

```
import React from 'react'
import ReactDOM from 'react-dom/client'
import App from './App'
import 'bootstrap/dist/css/bootstrap.min.css'
ReactDOM.createRoot(document.getElementById('root')).render(
  <App />
)
```

App.jsx:

```
import React from "react";
import { BrowserRouter, Routes, Route } from "react-router-dom";

import Navbar from "./Components/Navbar";
import Home from "./Pages/Home";
import Products from "./Pages/Products";
import Cart from "./Pages/Cart";
import "./App.css";
import { CartProvider } from "./Context/cartContext";

function App() {
  return (
    <CartProvider>
      <BrowserRouter>
```

```
      <Navbar />

      <Routes>
        <Route path="/" element={<Home />} />
        <Route path="/products" element={<Products />} />
        <Route path="/cart" element={<Cart />} />
      </Routes>

    </BrowserRouter>
  </CartProvider>
 );
}

export default App;
```

Navbar.jsx:

```
import React, { useContext } from "react";
import { Link } from "react-router-dom";
import { CartContext } from "../Context/cartContext";

const Navbar = () => {

 const { cart } = useContext(CartContext);

 return (
  <nav className="navbar navbar-dark bg-dark navbar-expand-lg">
   <div className="container">

     <Link to="/" className="navbar-brand">
       ShoppingCart
     </Link>

     <div className="ms-auto">

       <Link to="/" className="btn btn-outline-light me-2">
         Home
       </Link>

       <Link to="/products" className="btn btn-outline-light me-2">
```

```
        Products
       </Link>

       <Link to="/cart" className="btn btn-warning">
        Cart ({cart.length})
       </Link>

      </div>

    </div>
   </nav>
  );
};

export default Navbar;
```

ProductCardTemp.jsx:

```
import React, { useContext } from "react";
import { CartContext } from "../Context/cartContext";

const ProductCardTemp = ({ product }) => {

  const { addToCart } = useContext(CartContext);

  return (
   <div className="card shadow-sm h-100 product-card">

    <img
     src={product.image}
     className="card-img-top"
     style={{ height: "220px", objectFit: "cover" }}
    />

    <div className="card-body text-center">

     <h5 className="card-title">{product.name}</h5>

     <p className="text-muted">{product.brand}</p>

     <h6 className="text-success">₹{product.price}</h6>
```

```
      <button
        className="btn btn-primary mt-2"
        onClick={() => addToCart(product)}
      >
        Add To Cart
      </button>

    </div>

  </div>
 );
};

export default ProductCardTemp;
```

Home.jsx:

```
import React from "react";

const Home = () => {
 return (
  <div className="container mt-4 text-center">
   <h1>Welcome to Product Store</h1>
   <p>Select products and add them to cart.</p>
  </div>
 );
};

export default Home;
```

Products.jsx:

```
import React from "react";
import ProductCard from "../Components/ProductCardTemp";
import ProductData from "../Productdata";

const Products = () => {

 return (
  <div className="container mt-4">
```

```
    <h2 className="text-center mb-4">Products</h2>

    <div className="row">

      {ProductData.map((product) => (

        <div key={product.id} className="col-md-3 mb-4">

          <ProductCard product={product} />

        </div>

      ))}

    </div>

  </div>
 );
};

export default Products;
```

Cart.jsx:

```
import React, { useContext } from "react";
import { CartContext } from "../Context/cartContext";

const Cart = () => {

 const { cart, removeFromCart } = useContext(CartContext);

 return (
  <div className="container mt-4">

    <h2>Your Cart</h2>

    {cart.length === 0 ? (
      <p>No items in cart</p>
    ) : (

      cart.map((item) => (
        <div key={item.id} className="card p-3 mb-2 d-flex flex-row justify-content-between
align-items-center">
```

```
        <div>
          <h5>{item.name}</h5>
          <p>₹{item.price}</p>
        </div>

        <button
          className="btn btn-danger"
          onClick={() => removeFromCart(item.id)}
        >
          Remove
        </button>

      </div>
    ))

  )}

  </div>
 );
};

export default Cart;
```

CartContext.jsx:

```
import React, { createContext, useState } from "react";

export const CartContext = createContext();

export const CartProvider = ({ children }) => {

 const [cart, setCart] = useState([]);

 // Add product
 const addToCart = (product) => {
  setCart([...cart, product]);
 };

 // Remove product
 const removeFromCart = (id) => {
  const updatedCart = cart.filter((item) => item.id !== id);
  setCart(updatedCart);
 };
```

```
  return (
    <CartContext.Provider value={{ cart, addToCart, removeFromCart }}>
      {children}
    </CartContext.Provider>
  );
};
```

Productdata.jsx:

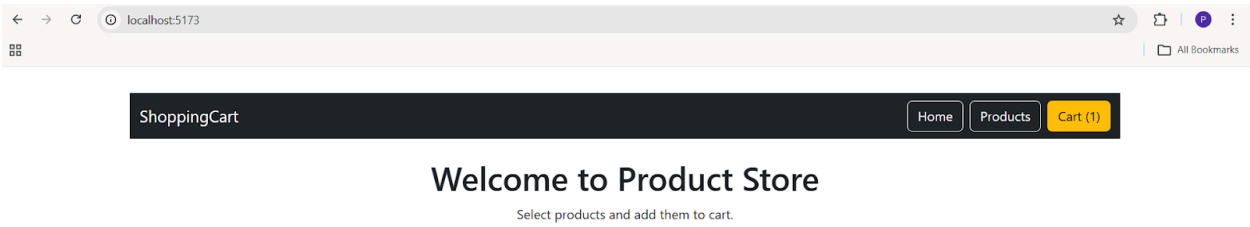
```
const ProductData = [
{
 id:1,
 name:"Iphone 14 Pro Max",
 brand:"Apple",
 price:120000
},
{
 id:2,
 name:"Galaxy S23",
 brand:"Samsung",
 price:90000
},
{
 id:3,
 name:"OnePlus 11",
 brand:"OnePlus",
 price:60000
}
]

export default ProductData
```

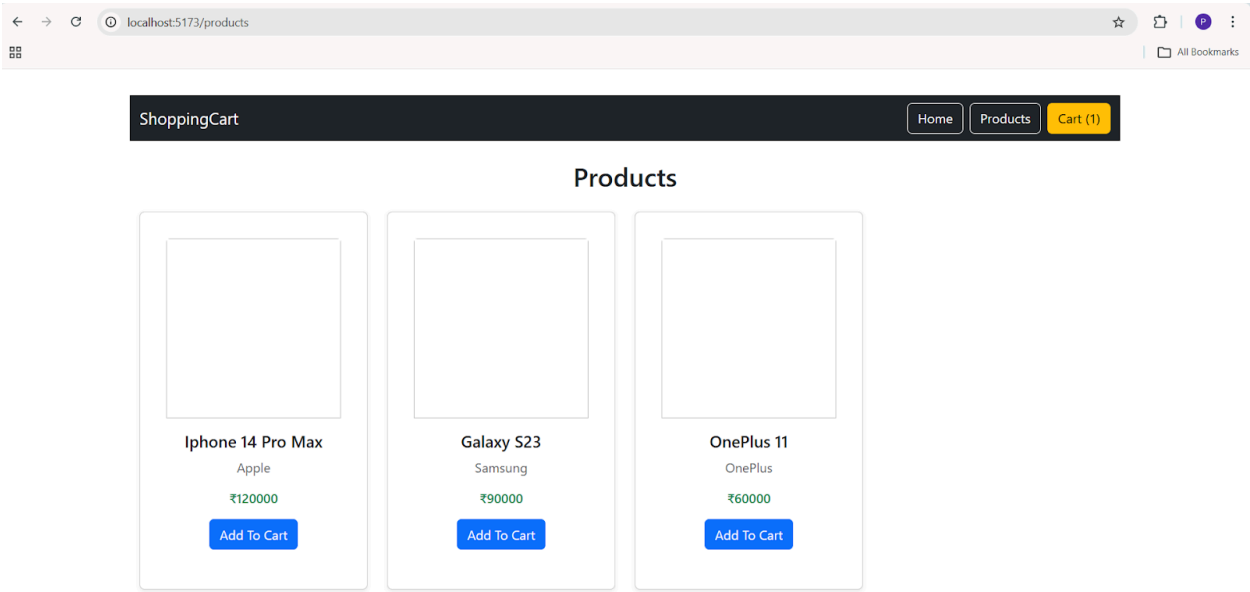
App.css:

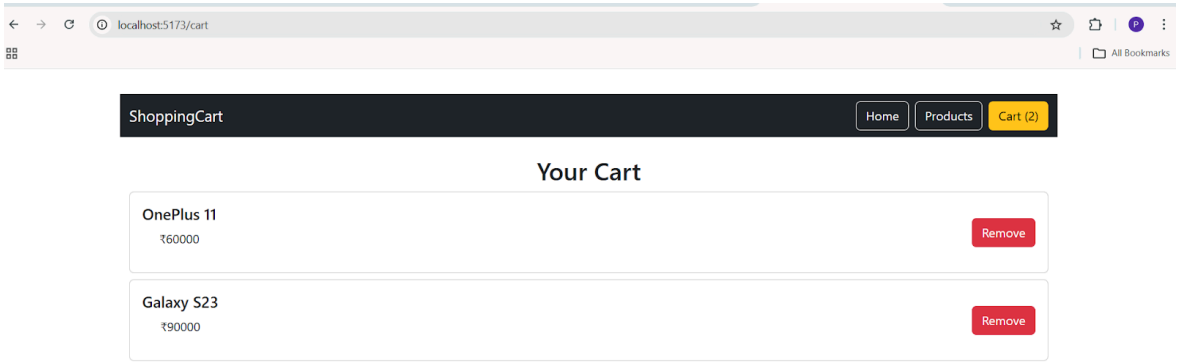
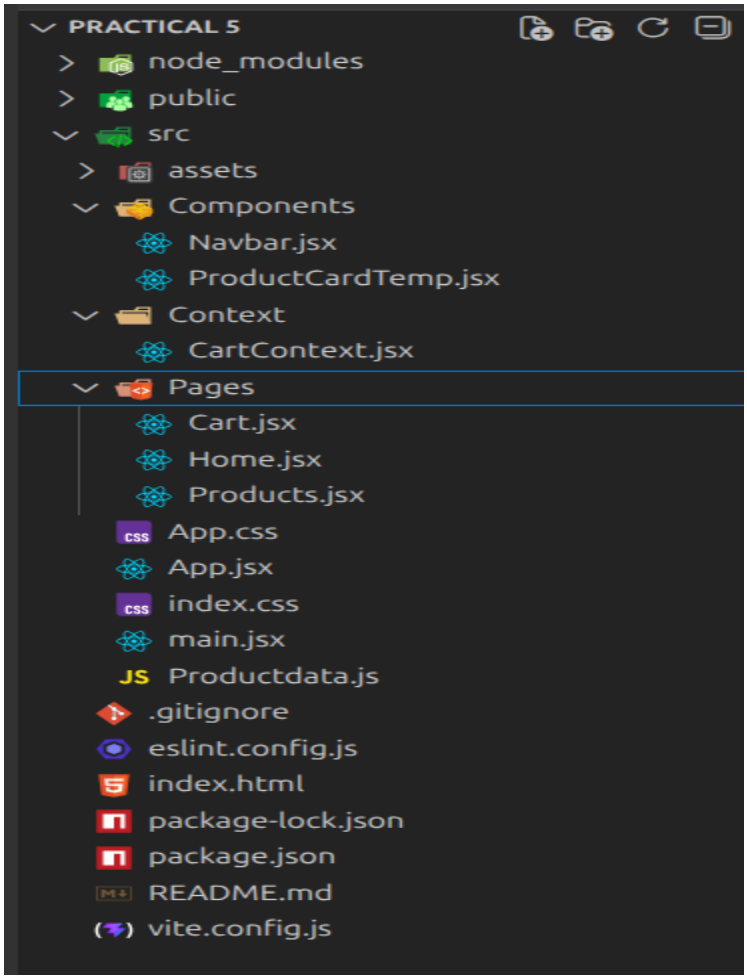
```
#root {
  max-width: 1280px;
  margin: 0 auto;
  padding: 2rem;
  text-align: center;
}

.logo {
```

```
  height: 6em;
  padding: 1.5em;
  will-change: filter;
  transition: filter 300ms;
}
.logo:hover {
  filter: drop-shadow(0 0 2em #646cffaa);
}
.logo.react:hover {
  filter: drop-shadow(0 0 2em #61dafbaa);
}

@keyframes logo-spin {
  from {
    transform: rotate(0deg);
  }
  to {
    transform: rotate(360deg);
  }
}

@media (prefers-reduced-motion: no-preference) {
  a:nth-of-type(2) .logo {
    animation: logo-spin infinite 20s linear;
  }
}

.card {
  padding: 2em;
}

.read-the-docs {
  color: #888;
}
.product-card{
transition: 0.3s;
}

.product-card:hover{
transform: scale(1.05);
}
```

Output:


localhost:5173
All Bookmarks
ShoppingCart
Home
Products
Cart (1)
Welcome to Product Store
Select products and add them to cart.


localhost:5173/products
All Bookmarks
ShoppingCart
Home
Products
Cart (1)
Products
Iphone 14 Pro Max
Apple
₹120000
Add To Cart
Galaxy S23
Samsung
₹90000
Add To Cart
OnePlus 11
OnePlus
₹60000
Add To Cart



File Structure: